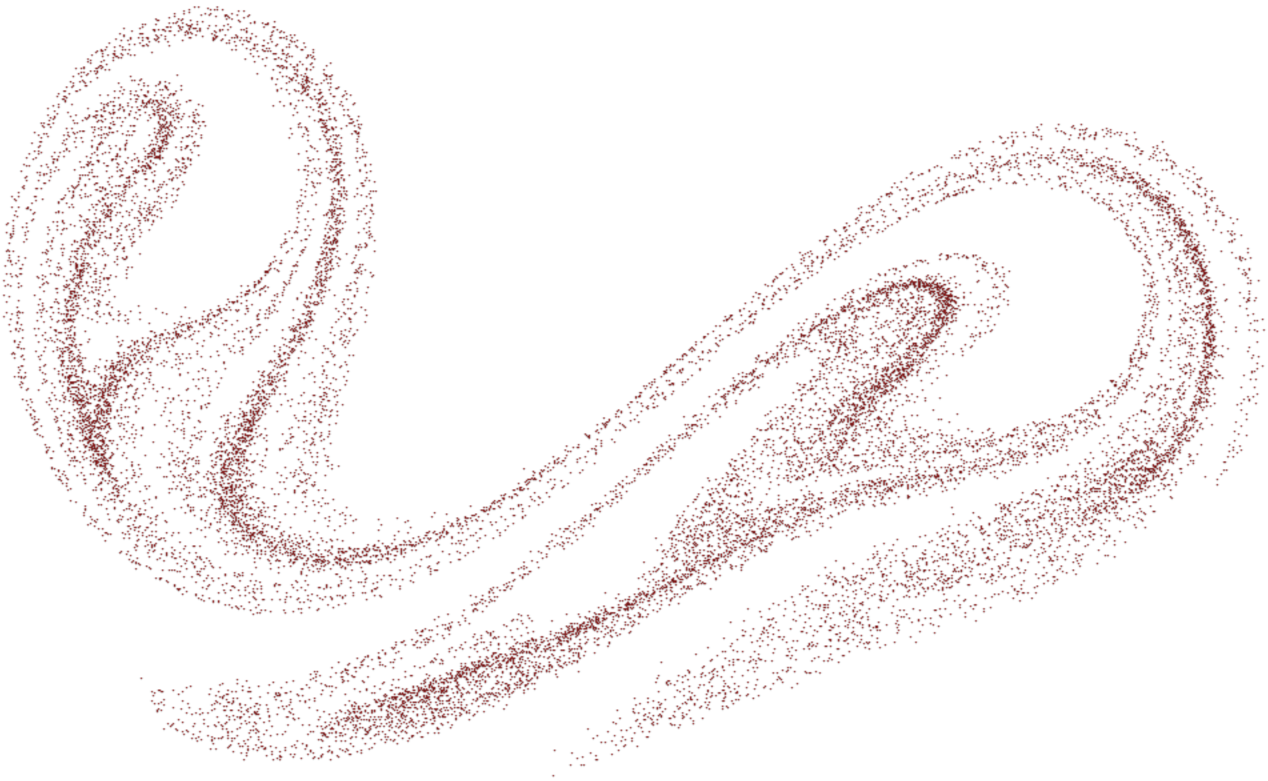
First year project

Chaos Theory



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1 Abstract

Non-linear dynamical systems can be highly sensitive to initial conditions, causing deterministic systems to develop complex, unpredictable behaviour. This project investigates the emergence of chaos through the numerical and experimental behaviour of a mechanical torsion pendulum. By deriving the non-linear differential equation governing the system, the study focuses on how variations in initial conditions and system parameters influence the transition from periodic to chaotic behaviour. This transition is used as the primary framework for inducing and demonstrating the presence of chaos in the system. Numerical methods, such as Bifurcation diagrams, Phase Plots and Poincaré sections, are used to analyse the dynamics of the system and demonstrate chaotic behaviour. These theoretical observations are confirmed through experimental analysis, demonstrating how subtle variations in the systems parameters can lead to unpredictable dynamics. The findings highlight the torsion pendulum as a clear example of a chaotic system, where deterministic equations of motion produce complex and unpredictable behaviour.

2 Use of generative artificial intelligence

The use of generative artificial intelligence in this report has been limited to assisting with selected parts of the MATLAB code, including assessment and refinement of the code. This entails verifying the accuracy of the code's theoretical implementation, debugging, and the generation of code snippets for numerical simulations and data analysis. All AI-generated content has been critically reviewed to ensure alignment with the established theoretical framework. This involved validating the code against theoretical predictions as well as experimental results, ensuring its accuracy and reliability. Further verification was achieved through guidance from supervisors and academic resources. Additionally, the AI grammar suggestions implemented in Overleaf were used to enhance grammatical accuracy and sentence structure.

Contents

3 Introduction

Commonly referred to as the 'butterfly effect', chaos theory is a mathematical framework that provides an understanding of non-linear systems, where precise laws do not guarantee predictable outcomes. Despite their deterministic nature, chaotic systems can exhibit complex and unpredictable dynamics due to a high sensitivity to initial conditions and system parameters. Understanding these principles is essential for applications in physics, engineering, and other fields where small variations can lead to significantly different outcomes. This project examines a mechanical torsion pendulum as an example to demonstrate these principles.

The torsion pendulum consists of a disk mounted onto a rod, free to rotate about its vertical axis, with restoring torque provided by springs attached to either side of the disk. While the system is governed by deterministic equations, its behaviour can transition from periodic to chaotic under certain conditions. By adjusting parameters such as the damping coefficient or the external driving frequency, it is possible to identify the precise threshold where the system's motion transitions from a stable periodic state into a chaotic one.

3.1 Objective

The objective of the project is to analyse the appearance of chaotic behaviour in a mechanical torsion pendulum, demonstrating the principles of chaos theory through theoretical modelling, numerical analysis, and experimental validation. This is accomplished by the following.

- Deriving the non-linear differential equation describing the system's motion.
- Implementing numerical approaches, such as bifurcation diagrams and poincaré plots, to analyse the transition from periodic to chaotic behaviour.
- Conducting experiments to validate theoretical predictions.
- Discussing the torsion pendulum as an example of deterministic chaos and its implications for understanding complex dynamical systems
- Drawing an analogy between the system and the heart as non-linear oscillatory systems.

3.2 Project boundaries

This project focuses exclusively on chaos in a torsion pendulum, examining the transition to chaotic behaviour. The purpose of this study is limited to comparing numerical results with experimental data, analysing how the system evolves from predictable motion to chaotic behaviour under specific conditions. The numerical code used is avaible on GitHub, [Cer26].

4 Equipment

To achieve this, the following setup was utilized. The torsion pendulum, as shown in figure 1, consists of three sensors, two springs, a disk, and a motor, all mounted in place using metal rods. To induce chaotic behaviour, a weight was added to the wheel, altering the dynamics of the system.

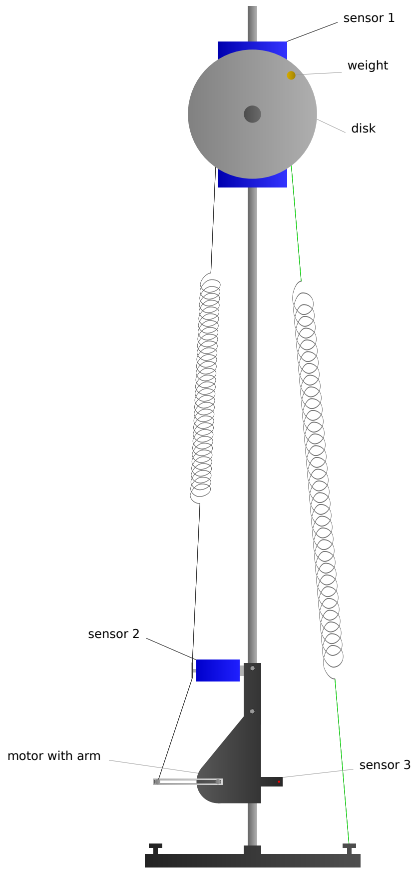

Figure 1: The torsion pendulum setup, including three sensors, two springs, a disk, and a motor, with added weight. The magnet, used for damping the disk, is fixed on the sensor 3 located behind the disk.

A magnet, positioned behind the disk on sensor 1, provides controlled damping to stabilize oscillations. In addition, sensor 3 was implemented to detect and record each completed period, ensuring accurate tracking of the pendulum movement. Pasco Capstone was used for data collection; it allowed for the tracking of the oscillations in real time. MATLAB was used to analyse, visualize and process data and numerical solutions.

5 Theory

To establish the theoretical framework, the mathematical expression for a standard torsion pendulum is first derived and then extended to include the elements that introduce chaotic behaviour.

5.1 Standard Torsion Pendulum

The dynamics of a standard torsion pendulum are governed by three torques, which is the rotational equivalent of force: a damping torque, which opposes the motion and dissipates energy; the restoring torque from the springs, which returns the disk to its equilibrium position; and the external driving torque provided by the motor, which introduces energy into the system. The damping torque in this system is proportional to the angular velocity because the magnetic damping mechanism generates a resistive force that increases linearly with the wheel's rotational speed. Applying Newton's second law for rotational motion, the net torque equals the system's inertial resistance to angular acceleration:

$$\tau_{damp} + \tau_{spring} + \tau_{outer} = I\theta''$$

The rotational version of Hooke's law states that the restoring torque is proportional to the angular displacement θ, rather than the linear displacement x, [Dav23].

$$F_{spring} = kx = k(r\theta)$$

The restoring torque is then derived from the force as follows.

$$\tau_{spring} = rF_{spring}\sin\left(\frac{\pi}{2}\right) = r[k(r\theta)] = \kappa\theta$$

Thus, the equation of motion for a standard torsion pendulum driven by a sinusoidal motor input is:

$$b\theta' + \kappa\theta + A\cos(\omega_e t) = I\theta''$$

5.2 Chaotic Torsion Pendulum

With the introduction of chaos, two key elements were added to the system, the torque of the weight and the non-linearity of the spring torque. The torque of the weight, τ_{weight}, depends on its mass m, the gravitational acceleration g, the distance d from the center of the wheel, and the component of the gravitational force that actively contributes to turning the wheel, which is proportional to $\sin(\theta)$, where θ is the angular displacement of the wheel. Gravity pulls the weight downward, and only the tangential component, which is parallel to the wheel's motion, affects the rotation, which is captured by the $\sin(\theta)$ term. This introduces a new equilibrium point for system, determined by the initial angular position of the weight, θ_0. By default, the weight is placed at the top of the wheel, where $\theta_0 = 0$. This causes a dynamic interaction between the spring and the weight, where the spring seeks to return the system to its natural resting position, while the gravity pulls the weight toward the point of its motion. Without the weight, the spring would naturally find equilibrium at a fixed angle. However, with the weight added, the system is driven to rotate until the weight reaches a stable position, creating a continuous interaction between the spring and the weight. If the weight is placed at an angle other than the top, this angle becomes the new θ_0, as illustrated in figure 2.

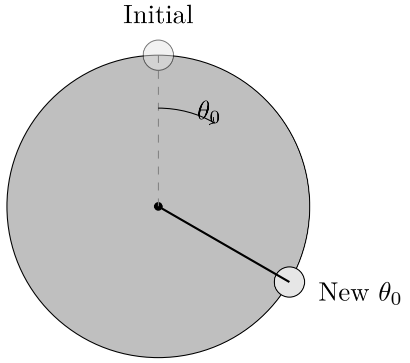

Figure 2: Disk with weight placed at an angle θ_0, with the initial position shown as a dashed line.

This shifts the natural resting position of the spring, altering the point of conflict between the spring's restoring force and the gravitational torque of the weight. Essentially, θ_0, defines where the system would naturally settle if left undisturbed, and dictates how the spring and weight interact. Thus, the equation of motion for the system becomes:

$$\tau_{damp} + \tau_{spring} + \tau_{weight} + \tau_{outer} = \tau_{res}$$

Substituting the expressions of the torques:

$$-b\theta' - \kappa(\theta - \theta_0) + mgd\sin(\theta) + A\cos(\omega_e t) = I\theta''$$

Expanding the spring term:

$$-b\theta' - \kappa\theta + \kappa\theta_0 + mgd\sin(\theta) + A\cos(\omega_e t) = I\theta''$$

This equation can be simplified by dividing through by the inertia I and rearranging the terms to isolate the external force term on the right-hand side.

$$\theta'' + \frac{b}{I}\theta' + \frac{\kappa}{I}\theta - \frac{\kappa}{I}\theta_0 - \frac{mgd}{I}\sin(\theta) = \frac{A}{I}\cos(\omega_e t)$$

The grouped constants can be replaced with single symbols for clarity, where δ represents damping, α is the stiffness of the system, the linear restoring force, β is the coefficient of the non-linear term enabling chaos, and γ is the amplitude of the external forcing:

$$\theta'' + \delta\theta' + \alpha\theta - \alpha\theta_0 - \beta\sin(\theta) = \gamma\cos(\omega_e t) \tag{1}$$

This results in a equation similar to the duffing equation with the difference being that β typically appears as the coefficient to θ^3, rather than $\sin(\theta)$, and θ_0 is not included as an additional term, as shown in eq. 2

$$x'' + \delta x' + \alpha x + \beta x^3 = \gamma\cos(\omega_e t) \tag{2}$$

However, this difference is not too important, as what matters the most is the presence of the non-linear term, which combined with the external energy and the right relationship between α and β, can induce chaotic behaviour, much like the Duffing equation. Chaos arises in this case through the emergence of a double well potential. For the equation of the torsion pendulum, the potential well can be derived from the potential energy of the system, which comes from the spring and gravitational forces. By integrating these forces over distance, we obtain the potential energy, since energy is the accumulation of force along a displacement:

$$\frac{U}{I} = \int \alpha\theta - \alpha\theta_0 - \beta\sin(\theta)\,d\theta = \frac{1}{2}\alpha\theta^2 - \alpha\theta_0\theta + \beta\cos(\theta)$$

The parameters of α and β are experimentally determined in section 6 as $\alpha = 7,971s^{-1}$ and $\beta = 47,73\frac{rad}{s^2}$. Using these values, the actual potential of the system can be modelled and represented accurately within the uncertainty and precision limits of the measurements, as depicted in figure 3.

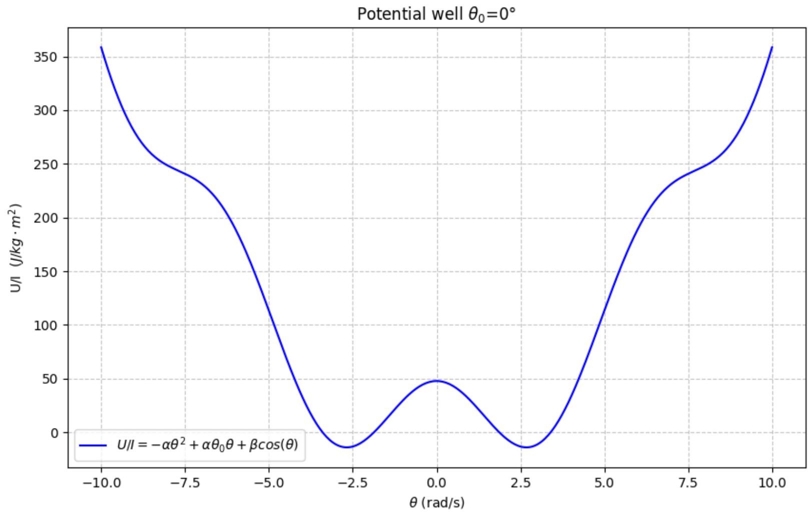

Figure 3: Potential double well for the system with $\theta_0 = 0°$, where the potential per inertia U/I, is presented as a function of the angular displacement θ. The graph reveals the two states which a Duffing-like chaotic system will switch between due to the opposing forces of springs and weight.

The double well features a critical point at the top of the hill between the two wells. In a periodic Duffing-like system, the system will oscillate back and forth within a single well. In a chaotic Duffing-like system, however, the oscillation is supplied with just enough energy to reach the critical points. From this point on, the system may either fall back into its original well or transition to the other well. The system will continuously reach this critical points, with the unpredictable outcome of falling into one well or the other. This makes the system extremely sensitive to initial conditions, as these determine whether the system will switch between wells at any given point in time. Although the outcome may appear random, it is actually a deterministic chaos. If the right conditions are not met, the double well may be uneven, making it harder to reach the critical points, and thus also more difficult to achieve chaos. By adjusting θ_0, an uneven double well can be created, demonstrating how certain conditions might prevent the system from behaving chaotically if not properly configured, as seen in figure 4.

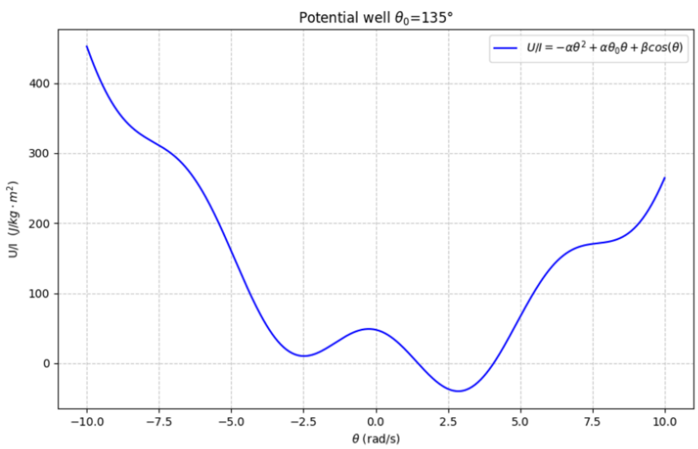

Figure 4: Uneven potential double well for the system with $\theta_0 = 135°$, where the potential per inertia U/I, is once again presented as a function of the angular displacement θ.

Unlike the symmetric double well seen in figure 4, this configuration introduces asymmetry that favours one of the two states. As a result, the system may become trapped in a single well, exhibiting periodic rather than chaotic behaviour. This demonstrates how changing θ_0 can prevent the system from achieving chaos.

5.3 Numerical solution of the mathematical equation

To analyse the system numerically, the Duffing-like equation is transformed into a set of first-order differential equations more suitable for computational solving. Instead of solving the equation in a single step, MATLAB approximates the solution by evaluating small differential increments and storing the resulting values in variables, (x_1, x_2, x_3), allowing the system to remember its current position. This transforms the numerical integration into a vector field described by the equations of the variables' change, (x_1', x_2', x_3'), showing how the system evolves over time

The first variable, x_1, is the angular displacement θ. The second variable, x_2, is the angular velocity $\omega = \theta'$, meaning it is the derivative of the first variable. The last variable, x_3, is time, t. The original system depends explicitly on time, making it non-autonomous. By defining $x_3 = t$, time is added as a state variable, which allows the system to be written off as an autonomous system that depends only on its variables.

$$\begin{aligned} x_1 &= \theta \\ x_2 &= x_1' = \theta' \\ x_3 &= t \end{aligned}$$

The second derivative equation, x_2', is the full Duffing-like equation because it is the second derivative of the angular displacement θ'', which can be isolated from the original equation.

$$\theta'' = -\delta\theta' - \alpha\theta + \alpha\theta_0 + \beta\sin(\theta) + \gamma\cos(\omega_e t)$$

Rewriting this in terms of the variables:

$$x_2' = -\delta x_2 - \alpha x_1 + \alpha\theta_0 + \beta\sin(x_1) + \gamma\cos(\omega_e x_3)$$

Letting t denote time in seconds, we find that the third derivative equation is $x_3' = 1$, indicating a constant rate of change.

6 Determining the parameters

In order to apply the theoretical framework, it is essential to have a precise determination of the system's parameters. After an extended period of trial and error, the system was optimized to create an even double well, enabling a 5,5-hour run that remained chaotic throughout. Alongside the experimental data from this run, a numerical model was analysed in order to demonstrate the accuracy of the Duffing like equation (1). Given the system's high sensitivity to initial conditions, the equation's constants were carefully calculated, especially inertia, which forms a part of most of the constant groups.

The inertia is derived from the disk and weights' composition and the weights' placement on the disk using the parallel axis theorem, which calculates an object's moment of inertia about any axis, provided it is parallel to an axis passing through the center of mass.

$$I = I_{wheel} + I_{weight} = 0.00016\ kg \cdot m^2 + 0.00015\ kg \cdot m^2 = 0.00031\ kg \cdot m^2$$

For the non-linear term β the remaining parameters were determined through direct measurements of the setup

$$\beta = \frac{mgd}{I} = \frac{0,015kg \cdot 9,82\frac{m}{s^2} \cdot 0,1m}{0.00031\ kg \cdot m^2} = 47.73\frac{rad}{s^2}$$

The stiffness α is derived from the stiffness of the springs, determined by applying a force and measuring the displacement.

$$\alpha = \frac{\kappa}{I} = \frac{0,00246\frac{N \cdot m}{rad}}{0.00031\ kg \cdot m^2} = 7.971s^{-2}$$

The damping factor δ is determined by the damping of the system, caused by the eddy current damping from the magnet. By performing a free oscillation and analysing the data using Pasco Capstone, the acquisition tool described in section 4, it is possible to estimate the damping factor B. This factor is related to the damping coefficient b as follows:

$$\delta = \frac{b}{I} = \frac{B2I}{I} = 2B = 2 \cdot 0,280\ s^{-1} = 0.56\ s^{-1}$$

In the original experimental setup, a small magnet positioned directly behind the upper motor was used to introduce damping. It was however later discovered that the system exhibited more distinct chaotic behaviour with lower damping. As a result, the magnet was placed further from the upper sensor, reducing the damping factor of the system.
The amplitude of the motor γ relates to the maximum power of the external driving source. It is calculated using the motor arm length L, the spring constant k, and the moment arm r of the pendulum.

$$\gamma = \frac{A}{I} = \frac{Fr}{I} = \frac{Lkr}{I} = \frac{0,085m \cdot 2,75\frac{N}{m} \cdot 0,025m}{0.00031\ kg \cdot m^2} = 18.92s^{-2}$$

Using the data from the free oscillation, the resonant frequency can be derived from the period of the oscillation. The resonant frequency is given by the following:

$$f_0 = \frac{1}{T} = \frac{1}{1.04} = 0.96s^{-1}$$

At this angular frequency, the system's energy input becomes more effective, allowing it to move past the critical point quickly. This frequency ensures optimal energy transfer for the system.
The corresponding angular frequency is:

$$\omega_e = 2\pi f_0 = 6.04\frac{rad}{s}$$

This represents the rate at which the system oscillates at resonance.

7 Results

Using the parameters calculated in the previous section, the system was analysed to understand its dynamic behaviour. To begin, a bifurcation diagram was generated to observe the conditions under which chaos occurs.

7.1 Bifurcation Diagram

The bifurcation diagram illustrates how the system's long term behaviour changes as the driving frequency ω_e is varied. Each vertical segment of the diagram corresponds to a specific driving frequency and the dots within the segment show the angular velocities, $\theta'_n(n = 1, 2, 3, ...)$, that the system lands on each period $T = \frac{2\pi}{\omega_e}$ once transients have been discarded. The individual dots indicate periodic behaviour, while the range of multiple dots within a vertical segment reveal chaotic behaviour, where the system examines a range of velocities without settling into a periodic pattern, as seen in figure 5.

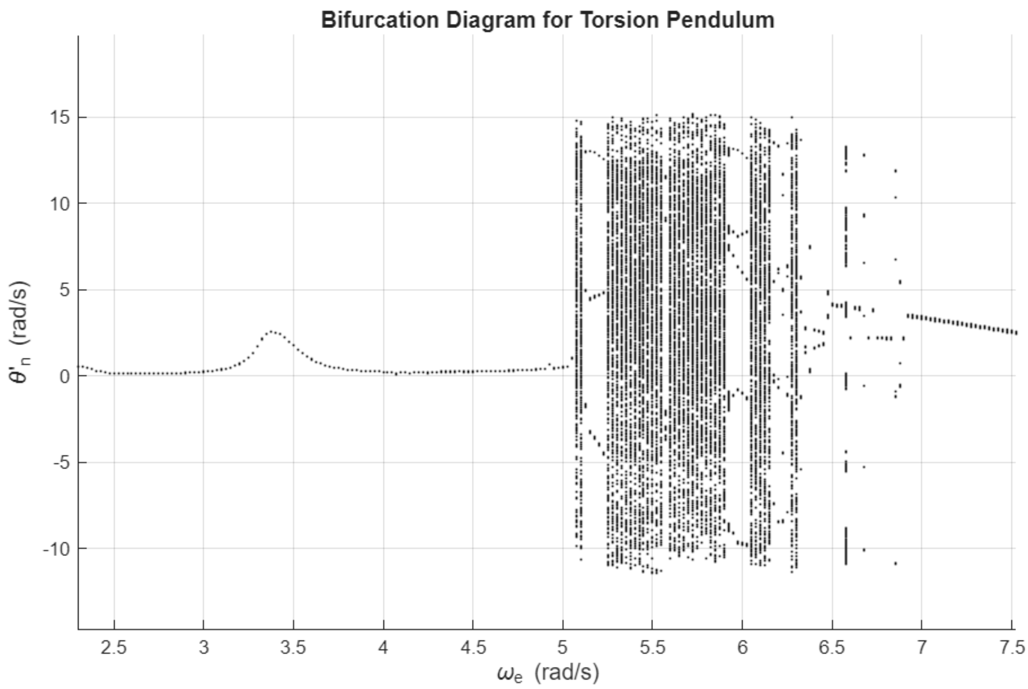

Figure 5: Bifurcation diagram showing the system's angular velocity $\theta'_n(n = 1, 2, 3, ...)$ as a function of driving frequency ω_e that the system lands on each period $T = \frac{2\pi}{\omega_e}$ once transients have been discarded. The individual dots in a vertical segment indicate periodic behaviour, while multiple dots in a vertical segment reveal chaotic behaviour.

To create this, a full simulation was conducted at a given driving frequency ω_e, with the period $T = \frac{2\pi}{\omega_e}$ used to measure the time between angular velocity measurements $\theta'_n(n = 1, 2, 3, ...)$. By running the simulation for 1000 seconds and removing the first 500 seconds of transients, it is possible to discard the transient phase, where the system appears chaotic, but does not represent the true asymptotic behaviour of the system, as seen in figure 5. The angular velocity values after each period were recorded in the plot. To ensure a smoother transition between simulations, the parameter continuation method was employed. This involves using the final angular velocity and displacement value from each run as the initial condition for the next run, incrementing the driving frequency by

$0.025\frac{rad}{s}$ each time. This approach ensures the transient period begins more smoothly, as the system is already near the parameters that previously produced a steady state. Furthermore, since the system could potentially settle into multiple periodic attractors depending on its starting conditions, the parameter continuation method increases the likelihood of finding the same or a closely related attractor. This provides a more systematic representation of the system's behaviour across different driving frequencies. By focusing on periodic behaviour within the same potential well, it becomes more clear when the chaotic behaviour arises as the points deviate from the line and begin to scatter around. As the driving frequency ω_e increases, the system appears to transition from a primarily periodic response, as seen by the individual points around a single angular velocity near $\theta'_n = 0$, into a regime suggesting multiple possible states of angular velocity. In the bifurcation diagram, this behaviour is reflected by the emergence of vertical segments of points, indicating that the system no longer converges to a single periodic orbit but instead explores multiple possible states depending on initial conditions or phase within the attractor. If the transient period had not been discarded, numerous additional points would appear above and below these individual points, incorrectly suggesting that the system had already begun period-doubling in response to the frequency input. Although this behaviour can be associated with period-doubling, where the oscillation period doubles as a control parameter is varied, the diagram itself does not provide sufficient evidence to confirm that a period-doubling cascade is occurring.

The observed structure in the bifurcation diagram is more consistently explained by a loss of stability in the single periodic attractor, leading to the emergence of chaotic dynamics starting near resonance $\omega_e = 6,04\frac{rad}{s}$. In this segment, the system is highly sensitive to initial conditions and phase, meaning that even minor differences in starting values can lead to distinct angular velocity for the same driving frequency. As a result, multiple asymptotic responses are captured at each parameter value, evident by the vertical spreading of points seen in the bifurcation diagram. Without the parameter continuation method, the singular points would exhibit abrupt jumps at certain frequencies, as the system might settle into unrelated fractal states. This would make the system appear less systematic. As the driving frequency approaches the resonance region, starting around $\omega_e = 5,1\frac{rad}{s}$, this effect becomes more pronounced, with a wider distribution of angular velocity values observed for each frequency step. This behaviour is characteristic of chaotic dynamics, where the system exhibits extreme sensitivity to initial conditions and explores a range of non-periodic states. The bifurcation diagram confirms that this resonance frequency is particularly effective for inducing chaos within the experimental setup. It is also evident that chaos emerges and vanishes abruptly at the boundaries between the vertical chaotic regions and the singular periodic points. This sharp transition happens because the system's trajectory begins to interact with a critical point in the potential double-well, [Hab23]. When the pendulum reaches the peak between the wells, the previously stable path within the original well becomes unstable. Even if the system returns to the old well, it may retain sufficient energy to reach the peak again, destabilizing the trajectory and leading to new, distinct paths. This repeated interaction with the critical point triggers an immediate transition into chaos, which is also known as a boundary crisis. The chaotic region spanning approximately in the range of $[-10, 15]$ in angular velocity, reflects a slight asymmetry in the potential wells. Despite the apparent randomness of chaotic behaviour, the system exhibits periodic windows, which are narrow ranges where the motion temporarily returns to a repeating pattern. In the pendulum's bifurcation diagram, notable periodic windows correspond to periods 3, 5, and 7, visible as distinct vertical clusters of points of the said number. This can be explained by Sharkovskii's Theorem, [Has11], which states that the presence of a period-3 orbit suggests the existence of orbits of all other integer periods. With period-3 being the highest in Sharkovskii's ordering, its presence implies the existence of infinitely many periodic behaviours within the chaotic regime. This suggests that with finer resolution and longer simulation times, the bifurcation diagram would reveal an infinite array of periodic behaviours fixed within the chaotic regime.

7.2 Time Series of Angular Velocity

By using the information gathered from the bifurcation diagram, which confirmed that a driving period of $\omega_e = 6.04\frac{rad}{s}$ would likely result in chaotic behaviour in the system, it was possible to conduct a run of 5,5 hours, as mentioned previously. The resulting time series of the angular velocity for both numerical and experimental data are presented in figure 6.

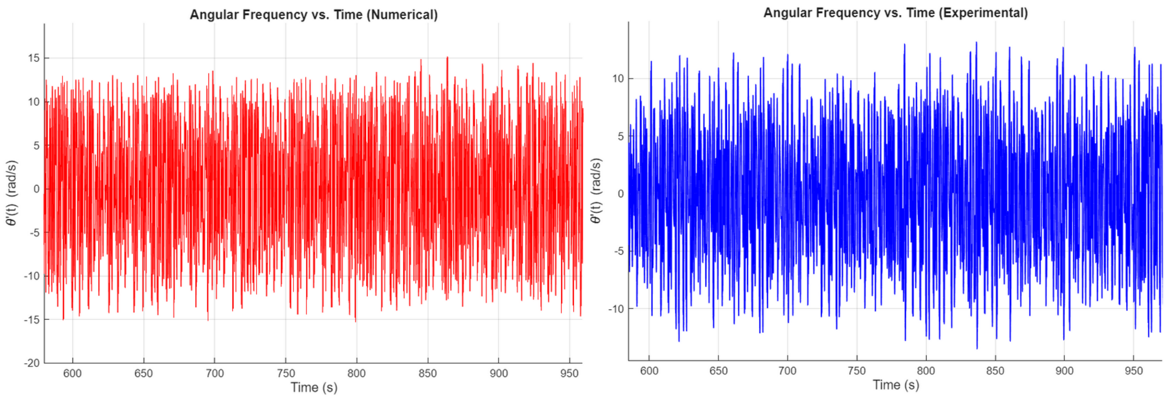

Figure 6: Time series of angular velocity showing how the velocity of the numerical solution (left) and experimental setup (right) changes over time.

The time series reveals irregular, non-repeating oscillations, as characterized by chaotic behaviour, which lasted throughout the entire run, suggesting the system to be chaotic. While a periodic system would exhibit repeating patterns, it is important to emphasize that these observations alone are not adequate to confirm chaos. The unpredictability of the time series could also appear from other complex or quasi-periodic dynamics. To more effectively determine whether the system is truly chaotic, additional analytical methods are essential. Power Spectral Density (PSD), Phase Portraits, Poincaré Plots, and fractal dimensional analysis are particularly useful for this purpose. These methods are able compress extensive data into a single diagram, revealing the system's true dynamical properties and providing the insight needed to distinguish true chaotic behaviour from other forms of complex motion.

7.3 Power Spectral Density (PSD)

An interesting model for analysing the behaviour of the system is the Power Spectral Density (PSD), which examines how power is distributed across the frequency spectrum. Unlike a periodic system which exhibits a distinct set of sharp, dominant frequencies, a chaotic system features a continuous, broad spectrum of frequencies due to its non-periodic and continuously changing nature. The PSD is derived from Fast Fourier Transformation (FFT), a technique that converts time-domain data, such as the pendulum's angular displacement $\theta(t)$, into frequency domain data $\theta(f)$. Directly applying a raw FFT to a large dataset of 400.000 data points would result in a highly noisy and impractically dense spectrum. To resolve this, the data was processed using Welch's method, which is a standard signal processing technique used to estimate the Power Spectral Density (PSD) of a signal, [Wu21]. Instead of analysing the entire dataset at once, Welch's method breaks the signal into smaller, more manageable parts, computing the spectrum for each, and then averaging them together. This averaging significantly reduces random fluctuations such as noise, making the structural peaks of the system clearly visible. To compute the PSD using the Welch method, the data is divided into overlapping windows of 128 actual data points, which were extended with zeroes to reach 1024 points, before applying the FFT. This improves the frequency resolution by effectively measuring the same 128 points over a larger range. The FFT separates the signal into its cosine and sine components, making the calculations more efficient, while also shifting each window by 64 points to ensure overlap, preventing data loss at the edges. The resulting FFTs were squared to derive the Power per frequency $P(f)$ and averaged in order to produce the final PSD. Due to a wide range of frequencies, the PSD is plotted on a base-10 logarithmic scale, showing $\log_{10}(P(f))$ vs. frequency, as seen in figure 7.

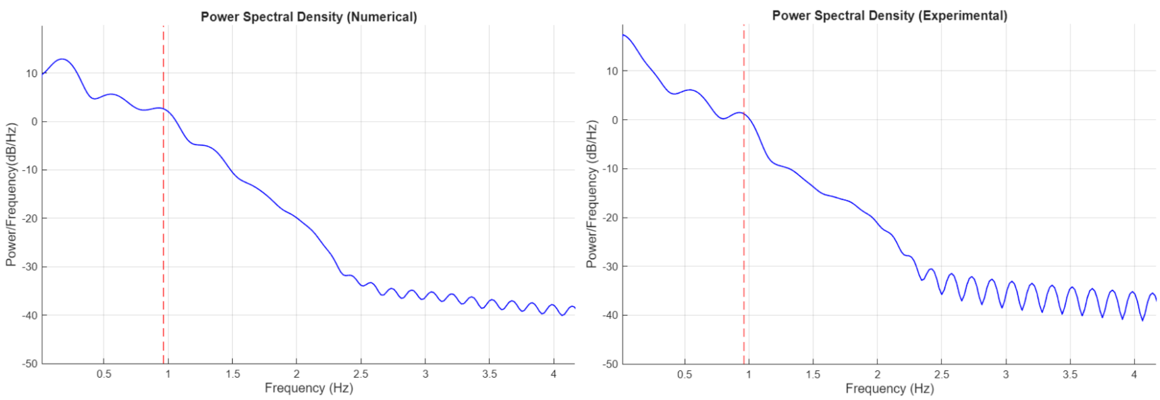

Figure 7: PSD for the numerical solution of the derived Duffing-like equation (1) (left) and the experimental data (right). The y-axis represents base-10 logarithmic scale of the power of each frequency, while the x-axis represents frequency in Hz. The red dotted line indicates the system's resonance frequency, f_0.

The PSD plots reveal a strong correspondence between the numerical and experimental models of the system, which confirms that the derived mathematical equation (1) accurately describes the pendulum's behaviour. Both spectrums exhibit a broad continuous decay in power with increasing frequency, rather than a few sharp peaks characteristic of periodic systems. This smooth distribution of power is an attribute of chaotic systems, where energy is spread across a wide frequency range due to the non-linear structure of the system rather than random interference. The peaks in the PSD correspond to different periodic trajectories, with the highest power spectral density occurring at very low frequencies, below the resonance frequency, which is marked by the red dotted line at $0,96$ Hz. This behaviour reflects the pendulum's transition between its two potential wells. Unlike a single well system, which would produce sharp peaks at constant frequencies, the chaotic behaviour of the system, which crosses the critical point at varying speeds, creates a smoother and more distributed energy spectrum. From the PSD, it was also possible to deduce that a sampling rate of 20 Hz would be sufficient as the dominant frequencies in the system lie below 10 Hz. As stated by the Nyquist-Shannon sampling theorem, the sampling frequency must be at least twice as high as the maximum frequency, [Tan05]. Therefore, a sampling rate of 20 Hz ensures that all significant frequencies are captured without any information loss and prevents aliasing.

7.4 Phase Portrait

While the PSD revealed the frequency characteristics of the system, a Phase Portrait provides another perspective by visualizing the system's trajectory in phase space. This is done by plotting the angular velocity $\omega(\theta, Z)$ as a function of the angular displacement θ and the motor contribution $Z = \gamma\cos(\omega_e t)$. By making the z-axis periodic rather than displaying time t, the portrait avoids an infinitely increasing axis. This is done by expressing the motor contribution $Z = \gamma\cos(\omega_e t)$ as a cosine function, causing it to oscillate between $-\gamma$ and γ. Since the cosine function is periodic, the z-axis repeats rather than increases indefinitely with time, which allows for a more intuitive comparison of the system's trajectories. Although a three-dimensional Phase Portrait provides a complete representation of the system's motion, its complexity can be difficult to interpret and is therefore often accompanied by a two-dimensional Phase Portrait, which omits the time dimension for clarity. Figure 8 depicts the three-dimensional Phase Portrait, while figure 9 presents the corresponding two-dimensional portraits.

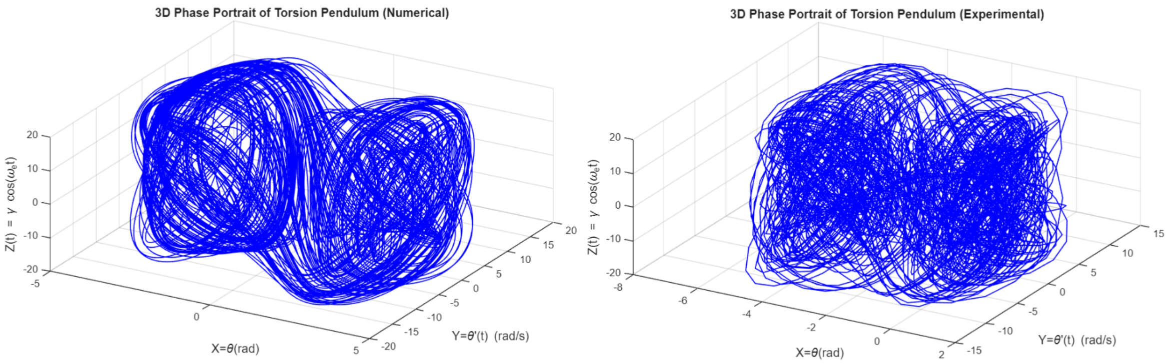

Figure 8: Three-dimensional Phase Portraits of the torsion pendulums trajectories in phase space, numerical solution on the left and experimental data on the right.

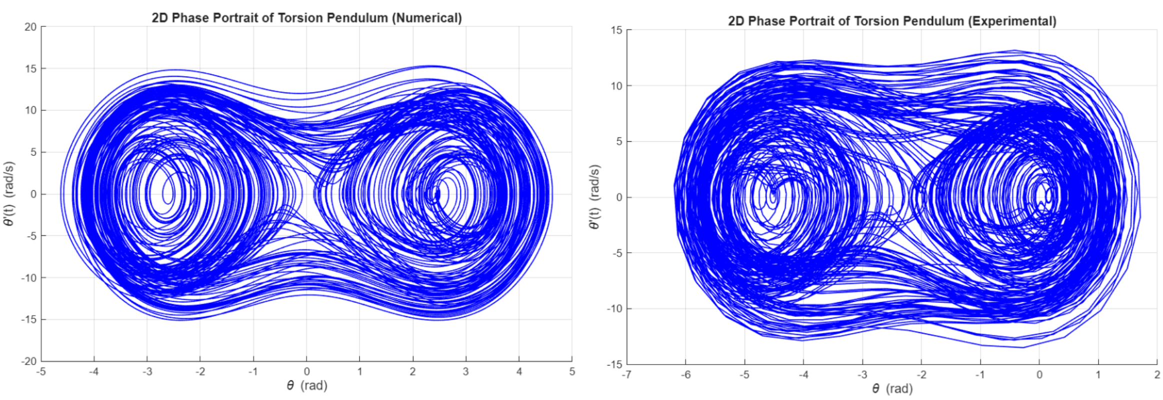

Figure 9: Two-dimensional Phase Portraits of the systems trajectories in phase space, showing angular velocity vs. angular displacement, numerical solution on the left and experimental data on the right.

In the three-dimensional Phase Portrait, it is apparent that the attractor never repeats itself, which is also a property of chaos, according to the Picard-Lindelöf theorem, [Sin13]. This theorem states that for differential equations, each initial condition, (θ, ω, Z), corresponds to exactly one unique solution. This means that if the system were to return to the exact same point, (θ, ω, Z), it would repeat its previous trajectory, resulting in simple periodic motions in the Phase Portrait. However, since the three-dimensional portraits only show the unique paths, they depict the system's chaotic behaviour. Both the three-dimensional and two-dimensional Phase Portraits reveal a strange attractor, a chaotic pattern that clearly moves around two equilibrium points corresponding to the system's two potential wells. The numerical and the experimental portraits share the same structure, though the experimental data exhibits slightly larger values, indicating minor discrepancies in the systems' damping. Additionally, the experimental portraits contain more noise, resulting in thinner lines at the edges. Both the experimental and numerical solutions display a subtle asymmetry, with the system spending slightly more time in the left potential well. Despite these minor differences, the numerical and experimental portraits are qualitatively the same, indicating that they represent the same dynamical system.

7.5 Poincaré Section

While the phase plots provide a visualization of the system's trajectory in phase space, they can be confusing and difficult to interpret, especially for chaotic systems. To simplify the analysis, a Poincaré section was implemented, which offers a clearer view of the system's dynamics by sampling the state of the system at fixed time intervals. It is central to chaos theory, serving as a fingerprint of the system's chaotic behaviour, and helps determine whether the mathematical equation and its derived values align with the experimental data. Similarly to the bifurcation diagram, the Poincaré section samples the angular displacement and velocity, (θ, θ'_n), at periodic intervals. In this analysis, the numerical solution uses slightly adjusted values to better match the experimental data while retaining the same model. Specifically, the damping δ is changed by $-0.2\,s^{-1}$ (35.71%), and the external angular velocity ω_e is adjusted by $0.09\ \frac{\text{rad}}{s}$ (1.49%). All other parameters remained unchanged, resulting in the graphs shown in figure 10, with a total adjustment of 6.2%.

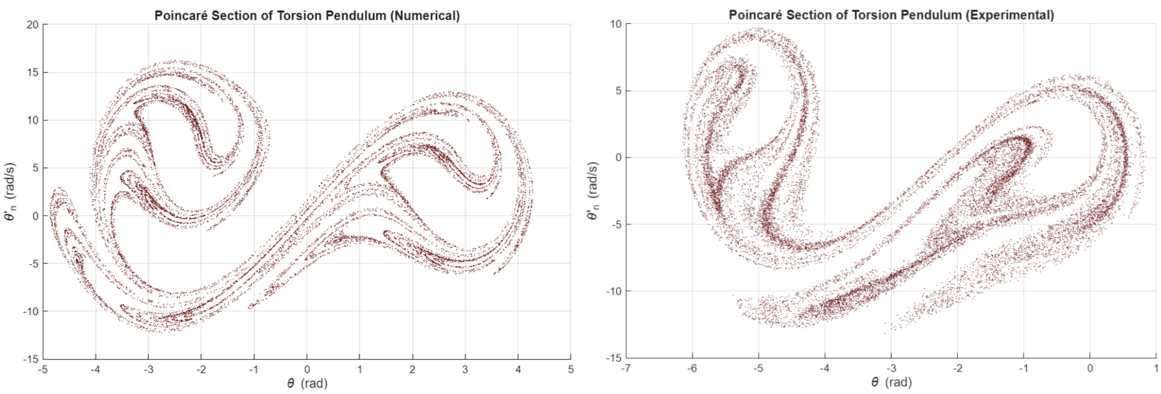

Figure 10: Poincaré section of the torsion pendulum, numerically solved (left) and experimental data (right), showing the system's angular displacement and velocity (θ, θ'_n) sampled at each period.

When comparing the Poincaré sections, both the numerical and experimental models reveal the same stretching and folding patterns, characteristic of a strange attractor. This attractor stretches toward two ends, as a result of the two potential wells, and forms a similar rounded shape in both models. The layers of lines in the attractors fold in the same places, with small differences in current values causing the system to range between wells and end in different states or layers. Due to the low damping and high angular velocity, ω_e, these layers exhibit visible gaps. The lack of damping and increased power allow the system to explore more states rather than being confined to a specific area. The similar stiffness in both models prevents excessive stretching, while the high non-linearity causes the ends to expand more vertically. Both models also display a tail beyond the two rounded ends. In the numerical model, the tail extends toward maximum velocity before dropping past zero, indicating a tendency for the system to stretch toward maximum angular displacement from the past equilibrium. The system then decelerates towards a value within this tail rather than moving directly to the rounded end. The experimental model, on the other hand, shows a tail where the system accelerates toward maximum velocity past equilibrium but falls into the tail's high displacement just before decelerating into the rounded ends. Despite these differences, both tails reveal a similar tendency, suggesting the same course into an uneven potential well. Both models also exhibit similar holds within the strange attractor, where the pendulum avoids certain areas despite being surrounded by them. This behaviour is due to islands of stability, where conditions at specific points in the period result in periodic oscillation. Topologically, the systems share the same general shape, though the experimental data includes more noise, separated points, and slightly different values due to the core property of chaos that small differences in initial conditions lead to significant differences in measured values. Initial conditions however, do not alter the overall shape, as demonstrated by the two models. The numerical and experimental models could be stretched into the same picture, indicating they are qualitatively identical from a topological standpoint. Thus, they are homeomorphisms, preserving the essential structure of the chaotic behaviour of the system.

7.6 Fractal Dimension

While the Poincaré sections depict a clear structure of the system's long-term dynamics, they do not directly quantify the complexity of the system. Fractal dimensions measure this structure more precisely. The fractal dimension is a measure of how a fractal pattern fills up the given space when observed at different scales. Unlike ordinary geometric objects, which have integer dimensions, such as a line that has one dimension or a surface which has two dimensions, fractals are made of structures that resemble collections of lines. These structures do not fully form a surface yet together exhibit a non-integer dimension. This indicates that they are more complex than a line, but they do not fully occupy a plane. In the context of a chaotic attractor, the fractal dimension describes how densely the points of the Poincaré section are distributed within the phase space. A higher fractal dimension corresponds to a more complex structure that occupies the phase space more effectively. This complexity arises from the repeated stretching and folding of trajectories in phase space, a characteristic of strange attractors, which are the fractal structures that emerge in chaotic systems. Their non-integer dimensions capture how the system's paths create complex, repeating patterns, which are deterministic in nature, yet sensitive to initial conditions. Nearby initial conditions remain uniquely determined by the system dynamics, but their trajectories can diverge significantly over time. As a result, the points in the Poincaré section form increasingly intricate structures when observed at finer scales.

A common method for estimating the fractal dimension is the box-counting method. The idea behind this method involves covering the phase space with a grid of boxes of size ε, and counting the number of boxes $N(\varepsilon)$ that contain at least one point of the attractor, [Tan20] . The total quantity of occupied boxes changes as the size of the boxes varies. The fractal dimension D is then estimated from the scaling behaviour of $N(\varepsilon)$ as $\varepsilon \to 0$:

$$D = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}$$

In practice, this relationship is evaluated using a log-log plot, where $\log N(\varepsilon)$ is plotted against $\log(1/\varepsilon)$. If the system exhibits fractal behaviour the data approximately follows a straight line over a range of scales, and the fractal dimension is obtained as the slope of this line. Using this method, the corresponding log–log plots for both numerical and experimental data are shown in figure 11.

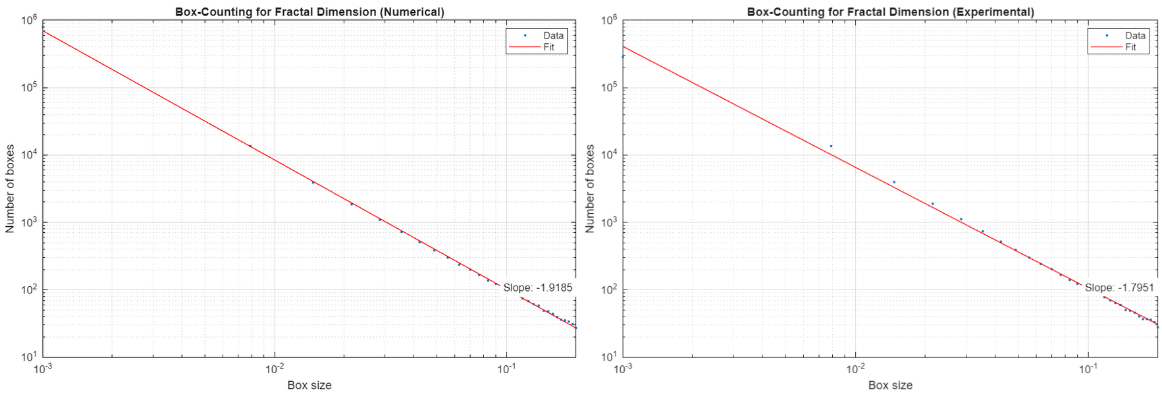

Figure 11: Log–log plot used for box-counting estimation of the fractal dimension. The number of occupied boxes $N(\varepsilon)$ is plotted against box size ε. The left panel (a) shows the numerical simulation, with a fractal dimension of $D = 1.9176$, while the right panel (b) shows the experimental data with a fractal dimension of $D = 1.7951$.

The fractal dimension using the box-counting method for the numerical solution shows a clear linear scaling region corresponding to a fractal dimension of $D = 1.9176$. The experimental data exhibits an increased scattering due to measurement noise and finite resolution, resulting in a slightly lower estimated fractal dimension of $D = 1.7951$. Both values lie between one and two, confirming that the attractor has a fractal structure that is more complex than a line but does not fully fill the plane. This discrepancy between numerical and experimental results highlights the sensitivity of fractal analysis to external disturbance, which can obscure the underlying deterministic chaos. Furthermore, the non-integer dimensions suggest that the dynamics of the system are governed by a strange attractor, which is a trademark of chaotic behaviour in non-linear systems.

7.7 Changing θ_0

As one of the final experimental analyses, the angular position θ_0 of the weight on the disk of the torsion pendulum was systematically varied through controlled experiments to examine its effect on the asymmetry of the system's potential well. As discussed in section 5.2, altering θ_0 changes the effective potential of the system, resulting in a progressively asymmetric potential well. This asymmetry is expected to affect the shape and distribution of trajectories in the phase portraits, reflecting the change in the underlying potential well. In order to examine this experimentally, the angular position θ_0 of the weight was gradually increased from $0°$, corresponding to the symmetric configuration of the system, in the clockwise direction. For each selected value of the angular position θ_0, a phase portrait was recorded while maintaining comparable initial energy conditions. This made it possible to compare how the system trajectories evolved as the potential well became increasingly asymmetric, as depicted in figure 12.

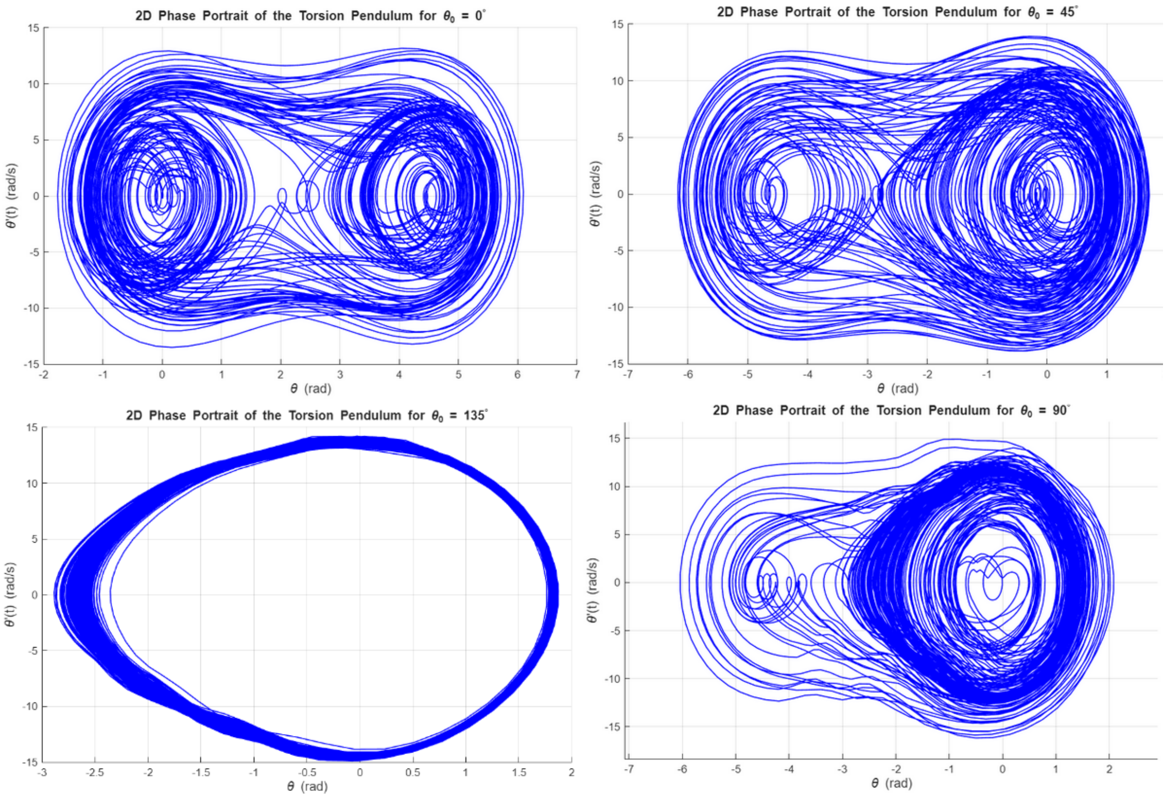

Figure 12: Two-dimensional phase portraits of the torsion pendulum as the potential well becomes increasingly asymmetric. The four portraits correspond to angular positions $\theta_0 = 0°, 45°, 90°, 135°$ arranged clockwise from the top-left corner, illustrating how system trajectories transform under comparable initial energy conditions.

The results demonstrate a clear transition in the pendulum's dynamical behaviour as the potential well asymmetry increases. At $\theta_0 = 0°$, the system exhibits relatively symmetric motion, with trajectories distributed around two comparable regions, indicating balanced oscillatory behaviour between the two potential wells. As θ_0 increases to $45°$ and $90°$, the symmetry diminishes and the trajectories become increasingly concentrated in one region, indicating that the motion becomes more uneven and strongly influenced by the asymmetry of the restoring torque. At $\theta_0 = 135°$, the behaviour changes more drastically, with the trajectories being confined in a singular large closed orbit within one of the two potential wells, suggesting that the system has transitioned to a motion centered around one stable region. Overall, the results show how changing θ_0 increases the asymmetry of the system, shifting the motion from nearly symmetric oscillations to more uneven and biased behaviour.

8 Discussion

Generally speaking, both numerical solutions and experimental analysis have consistently demonstrated fundamental characteristics of chaotic behaviour in the torsion pendulum. The bifurcation analysis revealed a transition from regular periodic oscillations to chaotic motion as the driving frequency approached the system's resonance region, indicating the onset of deterministic chaos in the torsion pendulum. To confirm that the observed behaviour was truly chaotic, several analysis methods were performed on both the numerical simulations and the experimental data. The Time series showed irregular, non-repeating oscillations, while the Power Spectral Density displayed a broadband of frequencies rather than distinct peaks, both characteristic of chaotic dynamics. The Phase Portraits further supported this behaviour, showing complex and irregular trajectories spread across phase space rather than the closed, repeating loops expected for periodic motion. Although the Poincaré plots were not fully comparable due to noise and experimental uncertainties, they still showed strong qualitative agreement with the numerical results. The presence of chaos was further supported by the box-counting method, which revealed a non-integer fractal dimension typical of chaotic systems. While the change in θ_0 did not directly demonstrate chaos, it depicted the system's sensitivity to parameter variations through changes in the in phase-space structure. Overall, the numerical and experimental analyses consistently indicated the presence of deterministic chaotic behaviour in the torsion pendulum. While deviations between the two were present, the numerical model reproduced the main non-linear features and overall dynamical behaviour observed experimentally.

8.1 Sources of error and measurement uncertainties

Due to the fact that chaotic systems are highly sensitive to initial conditions, even small experimental uncertainties could affect the motion of the torsion pendulum and contribute to discrepancies between the numerical and experimental results. Technical issues with the experimental set-up such as the sliding of strings off the disk, small shifts in the disk position, and slight vibrations from the table introduced disturbances into the system. In addition, parameters such as the initial angle θ_0, the alignment of the sensors and the positioning of the motor were adjusted manually, introducing additional uncertainty. An important factor would also be the damping coefficient b, which was particularly difficult to accurately determine. Small changes in the regression interval during free-oscillation measurements produced relatively large variations in the estimated damping value, resulting in a less precise damping parameter. This could be a possible explanation as to why the experimental Poincaré Section more accurately matched the numerical simulation when performed with a lower damping value. Similarly, the driving angular frequency could only be controlled indirectly through the input of the motor voltage in Pasco Capstone, meaning small voltage changes produced measurable differences in frequency. Fine-tuning the voltage showed that changes of approximately $0.01V - 0.03V$ could correspond to the same angular frequency according to Capstone. Beyond this range however, a change of $0.01V$ typically altered the angular frequency by roughly $0.08\frac{\text{rad}}{s}$. This suggests that differences between the numerical and experimental Poincaré Sections may partly be attributed to limitations in the precision of the Capstone measurement system. The inertia calculation introduced some uncertainty as well, since it assumes a perfectly uniform circular disk. In reality, the disk contains two small protrusions where the weight could be attached, making this approximation slightly less accurate.

8.2 Damping and Magnetic Field

One possible explanation for the differences between the numerical and experimental results would be the way damping was modelled in the system. The numerical model assumed a damping torque proportional to the angular velocity, $\tau_{damp} = b\theta'$. In the experimental set-up however, the magnet was placed as far away from the disk as possible in order to induce more chaotic behaviour, thereby reducing its effect on the system. Measurements showed that the damping obtained with the magnet present was within the uncertainty range of measurements performed without the magnet, as discussed in section 8.1, suggesting that the magnetic damping had a limited influence on the system. Further free-oscillation analysis indicated that the damping coefficient b was not constant. By performing free oscillations without the magnet in place and fitting a damped sinusoidal regression in Capstone, the damping coefficient could be determined in the same way as for the numerical model. Although damped sinusoidal regressions initially fit the data well, measurements taken at different maximum angular velocities θ'_{max} suggested that the effective damping varied slightly with the motion of the system. The results show that b is not constant, but instead appears to vary with angular velocity, approximately following a relation $b = k\theta'$, as seen in figure 13.

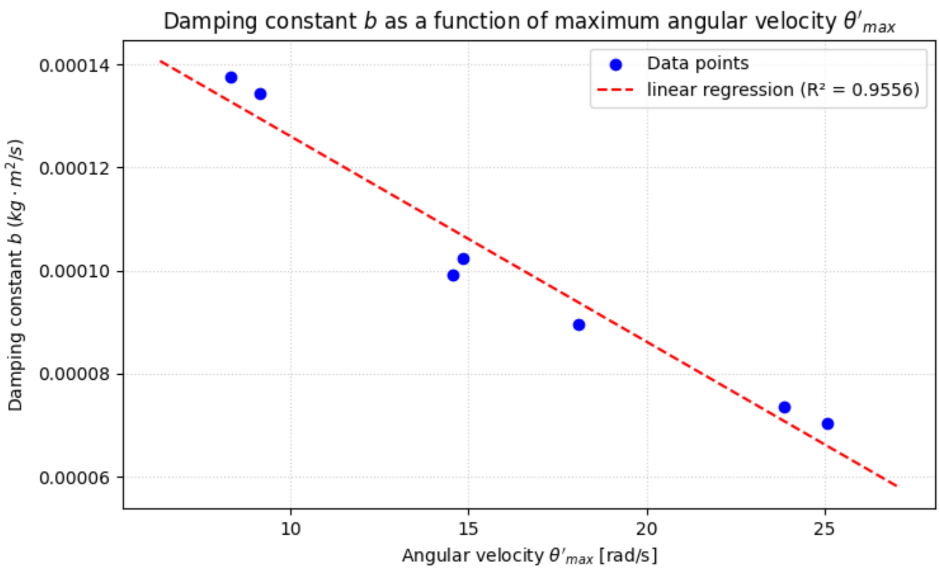

Figure 13: Relationship between the maximum angular velocity θ'_{max} of a free oscillation and the damping constant b. The measured data points (θ'_{max}, b) exhibit a strong linear correlation, as indicated by the regression fit $R^2 = 0.0556$, suggesting that the damping varies with angular velocity.

From these measurements, it can be concluded that the assumption of damping being solely proportional to angular velocity is only partially valid for the system. Furthermore, the damping decreases at higher speeds, indicating that the effect is not dominated by air resistance. Instead, it suggests a contribution from the Coulomb friction, which can influence the motion even though it does not depend on velocity. It is possible that a stronger presence of the magnet in the experiment would have had a greater effect on the observed chaotic behaviour, since the damping and velocity would show a linear relation. This could also have allowed the numerical solution to generate a more accurate representation of the system's behaviour. Despite these uncertainties, the numerical and experimental systems still showed strong qualitative agreement, suggesting that the model successfully captured the dominant non-linear and chaotic dynamics of the torsion pendulum.

9 Analogy

Although the torsion pendulum may seem far removed from our everyday lives, the characteristics of its motion are not unique to controlled laboratory systems. Similar patterns of variability and oscillation appear in many familiar phenomena. To better understand its dynamics, it is useful to compare it to a well-known oscillatory system, such as the heart. The heart does not beat with a single fixed period, instead its timing fluctuates in a structured manner due to the interaction of multiple regulatory feedback mechanisms, including the autonomic nervous system. This results in a heart rate variability, where the intervals between heartbeats, also known as RR-intervals, change over time rather than remaining constant. When these intervals are plotted as a histogram for a relaxed individual, the distribution shows a well-defined central tendency, forming a bell-shaped curve which reflects a preferred operating range rather than strict periodic behaviour. A similar type of analysis can be applied to the torsion pendulum by considering the time intervals between consecutive peaks in angular velocity. The pendulum produces a sequence of oscillations with peaks that can be interpreted as natural ‘beats’, analogous to the R-peaks in an electrocardiogram. Figure 14 shows the distribution of peak-to-peak time intervals obtained from the torsion pendulum experiment, alongside a representative RR-interval histogram from a heart rate variability study in a healthy individual, [Dav16].

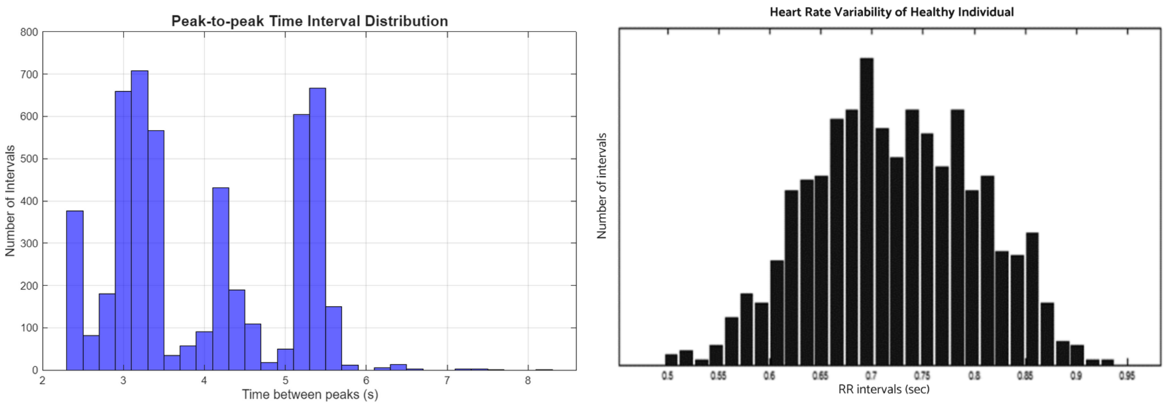

Figure 14: Histograms of peak-to-peak time interval distribution of torsion pendulum oscillations (left) and the RR-interval distribution associated with the heart rate variability in a healthy individual (right), [Dav16].

The right panel, representing the heart rate variability of a healthy individual, displays a broadly bell-shaped distribution of RR intervals centered around a dominant peak with moderate spreading, characteristic of healthy heart rate variability. In contrast, the left panel, corresponding to the torsion pendulum, exhibits a broader and more structured distribution with multiple peaks, indicating that the pendulum motion fluctuates between different preferred intervals. Despite the differences in shape, both distributions reflect systems that are not strictly periodic, but instead operate through structured variability rather than predictable recurrence.

A second perspective comes from examining the amplitude of each oscillation. While the interval distributions capture the variability in oscillation timing, the amplitude distributions reveal an underlying energetic stability. As shown in figure 15, the peak angular velocities of the torsion pendulum are centered around a dominant value, with a seemingly symmetric spread, closely resembling the distribution of R-peak amplitudes observed in a typical electrocardiogram signal.

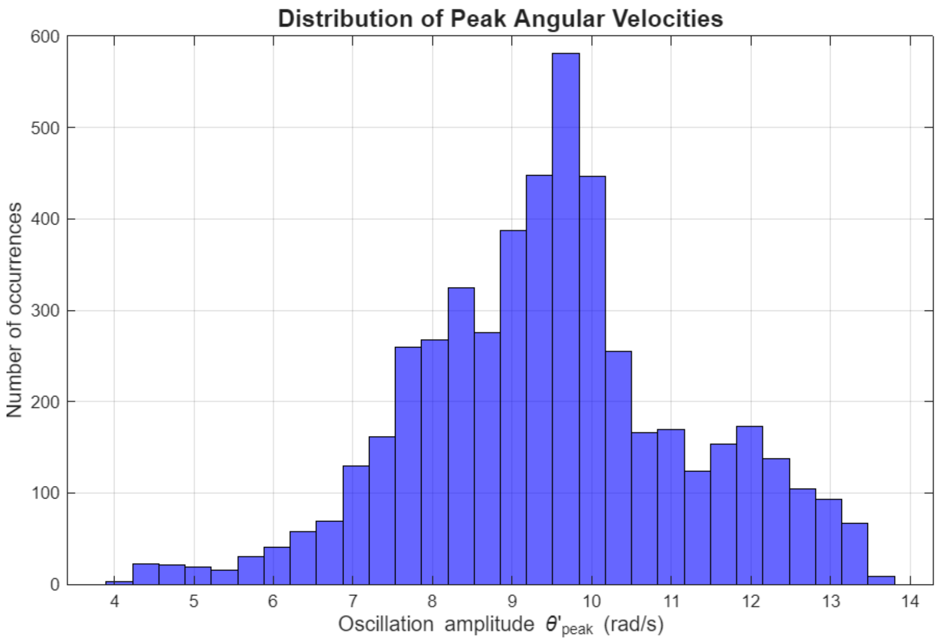

Figure 15: Histogram showing the distribution of angular velocities θ'_{peak} at oscillation peaks for the torsion pendulum.

This distribution confirms that, despite fluctuations in oscillation timing, the system maintains a relatively confined range of peak amplitudes. This indicates that the energy associated with each oscillation period remains consistent over time. In this regard, the torsion pendulum behaves analogously to the cardiac system, where R-peak amplitudes remain comparatively consistent. This stability in amplitude contrasts with the variability observed in the interval distributions discussed previously in the section, where there is no single dominant timing pattern. Instead, the system exhibits flexibility in the timing of oscillations while keeping the size of each cycle within a defined range. This discrepancy between timing variability and amplitude stability is a common trait of non-linear oscillatory systems, where different aspects of the dynamics are governed by distinct underlying mechanisms. Overall, these results show how the torsion pendulum and the heart can be interpreted within a common framework of structured variability in oscillatory dynamics.

10 Conclusion

In conclusion, both the numerical and experimental analyses consistently demonstrate that the torsion pendulum exhibits deterministic chaotic behaviour under the conditions studied in this report. The system showed clear signatures of chaos across multiple essential chaos theory tools, including bifurcation analysis, time series, power spectral density, phase portraits, poincaré sections, and fractal dimension. Main characteristics include sensitive dependence on initial conditions, broadband frequency content, and a fractal dimension structure. While some discrepancies were observed between the numerical and experimental results due to measurement uncertainties and modelling simplifications, the overall consensus in chaotic behaviour confirms that the numerical model successfully captures the main non-linear features of the system. Variations in system parameters further highlighted the sensitivity of the system's motion, supporting the chaotic nature of the system. Overall, the results provide strong evidence that the torsion pendulum behaves as a chaotic non-linear system near resonance, governed by deterministic but highly sensitive dynamics.

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